

Memory design has to follow certain guidelines in order to comply with the standards. While this helps with efficiency of the memory, it restricts design. According to Singh, "Integration will be mild since it has to abide by the standard form factor laid by the associations."

A look at the developments

We can all accept the fact that NAND flash came as the turning point for memory. Since then it has seen many changes, and today we have 512GB in a micro-SD-size form factor. Such additions and changes in the basic structure of memory have made it more lucrative and affordable for the user.

Popularisation of SSD. With the commercialisation of SSDs, there has been a rapid change in the way our devices work. "When we talk about SSDs, previously there was only SLC technology that was primarily targeted at the enterprise sector. It is further being evolved into MLCs, eMLCs, TLC, 3D TLC NAND and now M.2 form factor based SSDs targeted at both commercial and civil sector," says Prio Samuel, senior consultant global electronics, 6wresearch.

On-the-go connectivity. With consumer products being in the limelight, one technology making rounds is on-the-go (OTG) connectivity. Due to its ease of connectivity with smartphones as well as computers, it has become a handy tool with more products being launched in this segment.

Kingston DataTraveler MicroDuo 3.0 and SanDisk iXpand flash drives are some products that allow a user to access data to and from smartphones and tablets. OTG compatibility makes these devices versatile and allows a user to free memory in urgency.

Automatic data correction.

Among several methods that make the memory smart and avoid unwanted bit-flips, immunity-aware programming and error correction

The perfect storage

Some technologists describe the perfect storage as cheap, fast and non-volatile. Any memory that could fit into these parameters would be the holy grail of memory.

Claims have been made for 3D Xpoint to be ten times the density of today's DRAMs and 1000 times faster than traditional storage. This could bring about a change in traditional systems by reducing multiple layers of memory hierarchy and connecting these down to essentially just one layer of 3D Xpoint memory pool.

codes (ECCs) have come up. "With evolving NAND manufacturing processes, ECC is becoming notably more sophisticated, imposing heavier burdens on the host processor, especially for large-capacity NAND," says Kumar. ECC memory can detect and correct most type of internal data corruption.

Easier access to the processor (cache-focused design). Read only memory (ROM) is for permanent storage of data within a system on chip (SoC) or other integrated circuit applications. ARM offers both via programmable ROM and diffusion programmable ROM, delivering maximum flexibility in design application. High-density and low-power ROM memories are suitable for BOOT code.

More size in less space. Access time is directly related to the distance from the processor. In order to reduce this access time/distance from the processor, the memory module has to be designed under certain constraints. There is only so much memory you can fit into a memory module, while keeping it close to the processor. This calls for integrating more memory into lesser space.

Accessing 3D. Once you reach a strong foundation, there is nowhere to go but up. Keeping this in mind, we have memory concepts trying to access the third dimension (3D) and keep the memory closer to the processor in order to increase data-processing speeds. Singh makes it simpler to understand with, "3D NAND takes today's flash, which is built on a horizontal plane, and turns it sideways."

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Moving to 3D allows for a higher bit density in a smaller footprint while eliminating the limitation on lithography patterning.

Challenges still. SSDs are taking over the market, and very soon we might see the traditional magnetic disk only in old servers or as images. However, cost management is still a factor. "Due to high cost constraints of SLCs and MLC SSDs, end users from both commercial and residential sector prefer hybrid HDDs over traditional HDDs, along with low-cost TLC SSDs," explains Samuel.

Storage wars

"With flash technology advancement from the beginning of SLCs, MLCs to TLCs and now 3D NAND, greater capacity flash storage would be a norm in the near future," says Singh.

Stored data is constantly increasing, and by some estimates, the indexed Internet data will cross 1000 yottabytes (YB) very soon. This calls for similar storage options. Today, we have SSD memories that are compact, can carry higher data and are faster than traditional magnetic memories. Today is an era of information, and the highest storage options have to allow better analytics and success. **EFY**